

xTern: Energy-Efficient Ternary Neural Network Inference on RISC-V-Based Edge Systems

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Abstract—Ternary neural networks (TNNs) offer a superior accuracy-energy trade-off compared to binary neural networks. However, until now, they have required specialized accelerators to realize their efficiency potential, which has hindered widespread adoption. To address this, we present xTern, a lightweight extension of the RISC-V instruction set architecture (ISA) targeted at accelerating TNN inference on general-purpose cores. To complement the ISA extension, we developed a set of optimized kernels leveraging xTern, achieving 67 % higher throughput than their 2-bit equivalents. Power consumption is only marginally increased by 5.2 %, resulting in an energy efficiency improvement by 57.1 %. We demonstrate that the proposed xTern extension, integrated into an octa-core compute cluster, incurs a minimal silicon area overhead of 0.9 % with no impact on timing. In end-to-end benchmarks, we demonstrate that xTern enables the deployment of TNNs achieving up to 1.6 percentage points higher CIFAR-10 classification accuracy than 2-bit networks at equal inference latency. Our results show that xTern enables RISC-V-based ultra-low-power edge AI platforms to benefit from the efficiency potential of TNNs.

I. INTRODUCTION

Edge systems such as Internet of Things (IoT) sensor nodes are producing ever-increasing amounts of data. Simultaneously, machine learning (ML) techniques have revolutionized the state of the art in interpreting such data, from computer vision to medical data processing. To decrease latency and communication energy, alleviate privacy concerns, and improve the versatility of sensing systems, the trend has been to move the ML-based processing of sensor data away from centralized cloud servers and onto the device that collects them, a paradigm termed “edge AI”. Thanks to their ubiquity and low cost as well as their low operating power, microcontroller (MCU)-based systems are one of the most popular targets for the deployment of edge AI applications [1]. However, MCUs also come with stringent memory and processing resource constraints, making the deployment of large, compute-intensive deep neural networks (DNNs) to these platforms highly challenging and elevating the importance of careful optimization of ML algorithms to the target hardware.

Quantization has proven an essential technique to optimize the efficiency of DNNs for edge deployment. By representing network parameters and intermediate activations as low-bitwidth integers rather than full-precision floating-point numbers, a model’s compute resource, storage, and memory

requirements can be decreased. Together with innovations in efficient model topologies [2], [3], research in quantization has enabled models running on sub-100 mW MCU platforms to achieve statistical performance that would have required high-performance workstations less than a decade ago [4]. Given a suitable hardware platform, the most aggressive quantization schemes offer the largest efficiency gains: In binary neural networks (BNNs), parameters and activations take values in $\{-1, 1\}$ and are represented by a single bit, and in ternary neural networks (TNNs), they may additionally take the zero value. In the domain of extreme quantization, research on TNNs has shown that moving from binary to ternary quantization increases statistical accuracy significantly and results in superior accuracy-complexity trade-offs [5]. We illustrate this with an example in Section II-A.

In the quest for efficient inference, dedicated hardware accelerators for quantized neural networks (QNNs) of all forms, from BNNs to 8-bit and mixed-precision networks, have seen success in recent years [6], [7]. However, application-specific accelerators imply a large commitment in silicon area dedicated solely to QNN inference, which may be unacceptable for low-cost embedded systems. In such cases, implementing hardware support for low-precision arithmetic in general-purpose processing cores with custom instruction set architecture (ISA) extensions offers an attractive alternative to single-purpose accelerators. Although operational efficiency is generally lower, implementation overhead is greatly decreased, and efficiency gains over the unmodified system are substantial. ISA extensions also provide flexibility – while accelerators commonly support a finite set of layers, well-designed low-precision arithmetic instructions can be used to implement arbitrary kernels. For example, recent successes in weight ternarization of large language model (LLM) architectures [8] suggest that fully ternarized transformers may enable LLM inference on edge systems; ISA-level support for ternary arithmetic will prepare systems for these future architectures. Researchers have proposed several ISA extensions for low- and mixed-precision integer arithmetic and used them to optimize the accuracy-energy trade-off. However, while integer-bitwidth low-precision operations readily map to single instruction, multiple data (SIMD)-based ISA extensions, efficient execution of TNNs has thus far been restricted to dedicated accelerators due to their non-power-of-two value range.

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